Discount Guardrails: Where Deep Discounts Erode Profit

Report for the Head of Sales · US office-supplies retailer, orders 2014–2017

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About this report. This is an independent portfolio project. The data is a public sample retail dataset (US office supplies, 2014–2017, 9,994 order lines). The Head of Sales, who is setting next year's discount policy, is simulated, and none of the recommendations have been put into practice. Every number comes from a file in `/results`, which the SQL in `/sql` produces.

1. Summary

The decision is what maximum discount each product sub-category should allow.

My recommendation is a 20% cap, with manager approval for exceptions, on the eight sub-categories that were discounted above 20% and lost money on it. Tables, Storage and Supplies lost money even at 20% off, so their prices need a separate review. I would test the cap in a two-quarter pilot before rolling it out.

What the data shows:

- Order lines discounted above 20% were 13.9% of lines and 15.8% of sales, and they lost **$135,376**. Total profit was $286,397. All eight discount levels above 20% made a net loss.
- The discount is the cause. At no discount, the products sold at 30–50% off would have earned a 26.6% margin. They earned −19.7%.
- This wasn't a handful of bad deals. 565 of 793 customers (71%) received at least one discount above 20%. The loss is more concentrated by product: 27 products account for half of it.
- The pattern held in every year, segment and region.

What a cap might have been worth is a modelled figure, not a measured one. Re-pricing the 2014–17 orders at a 20% cap changes profit by **+$135k to +$219k**, depending on how much of the capped volume customers would still buy. If every affected order were lost outright, the change is still **+$111k**. The cap only loses money if it drives customers away. Those 565 customers made $299,733 on their other purchases, so losing more than about 45% of that would wipe out the gain.

The biggest unknown is how customers react to smaller discounts, and why each discount was given in the first place (clearing old stock, for example). The data can't answer either. The pilot can.

2. Context and scope

The commercial team sets discount limits once a year and checks them each quarter. I broke the decision into five questions:

#	Question	Decision it informs
Q1	Where do discounts line up with losses?	Which sub-categories need a cap
Q2	Is it the discount, or the products that get discounted?	The cap level for each sub-category
Q3	How concentrated are the losses?	A blanket rule or case-by-case approval
Q4	What would different caps have meant for 2014–17?	Which cap to pilot
Q5	Does the pattern hold across years, segments and regions?	One policy everywhere, and how to monitor it

I didn't look at sales-rep performance (the rep field looks randomly assigned, see section 6), customer retention, price elasticity or forecasting.

3. Data and method

The data has four tables. ORDERS has one row per order line: 9,994 rows covering 5,009 orders. PRODUCT has 1,862 products, CUSTOMER has 793 customers with one listed location each, and EMPLOYEES has 9 reps, which I didn't use. Every key matches.

All the analysis runs off one view, `fct_order_line`. It joins each order line to its product and customer without duplicating anything (9,994 rows in, 9,994 out). The source PRODUCT columns are mislabelled, so the view renames them: `NAME` → category, `CATEGORY` → sub-category, `SUBCATEGORY` → product name.

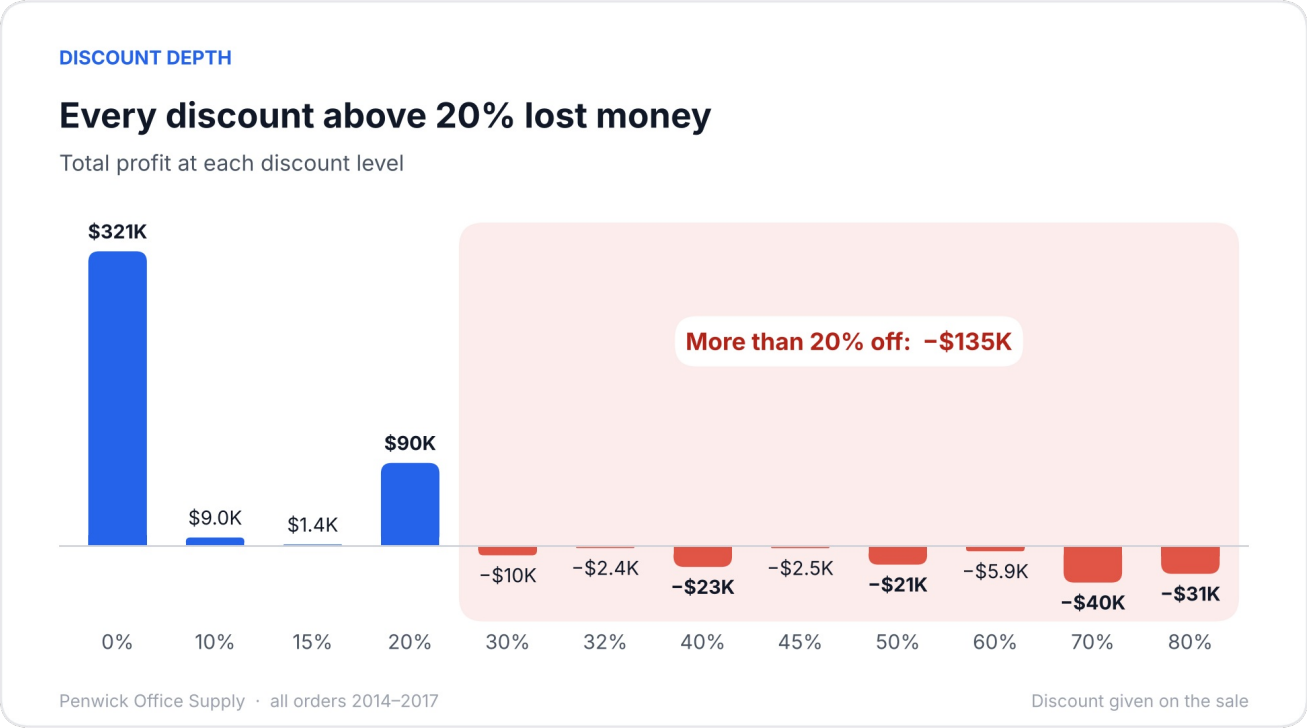
There are 12 discount levels in the data, from 0% to 80%. I grouped them into four bands: No Discount (0%), Standard (over 0% up to 20%), Deep (over 20% up to 50%) and Clearance (over 50%). The levels actually used above 20% are 30–50% (Deep) and 60–80% (Clearance), so the charts label the bands that way.

Margin is SUM(profit) ÷ SUM(sales), always a ratio of totals and never an average of line margins. Sales are after discount. The data gives profit but no cost, so I worked out an implied cost as sales − profit.

I ran 33 checks and all of them pass. They cover row counts, keys, joins, reconciled totals, dates, the what-if arithmetic, and a second calculation of the main tables in pandas. The full list is in `results/validation_log.csv`.

4. Findings

Q1. Every discount level above 20% lost money

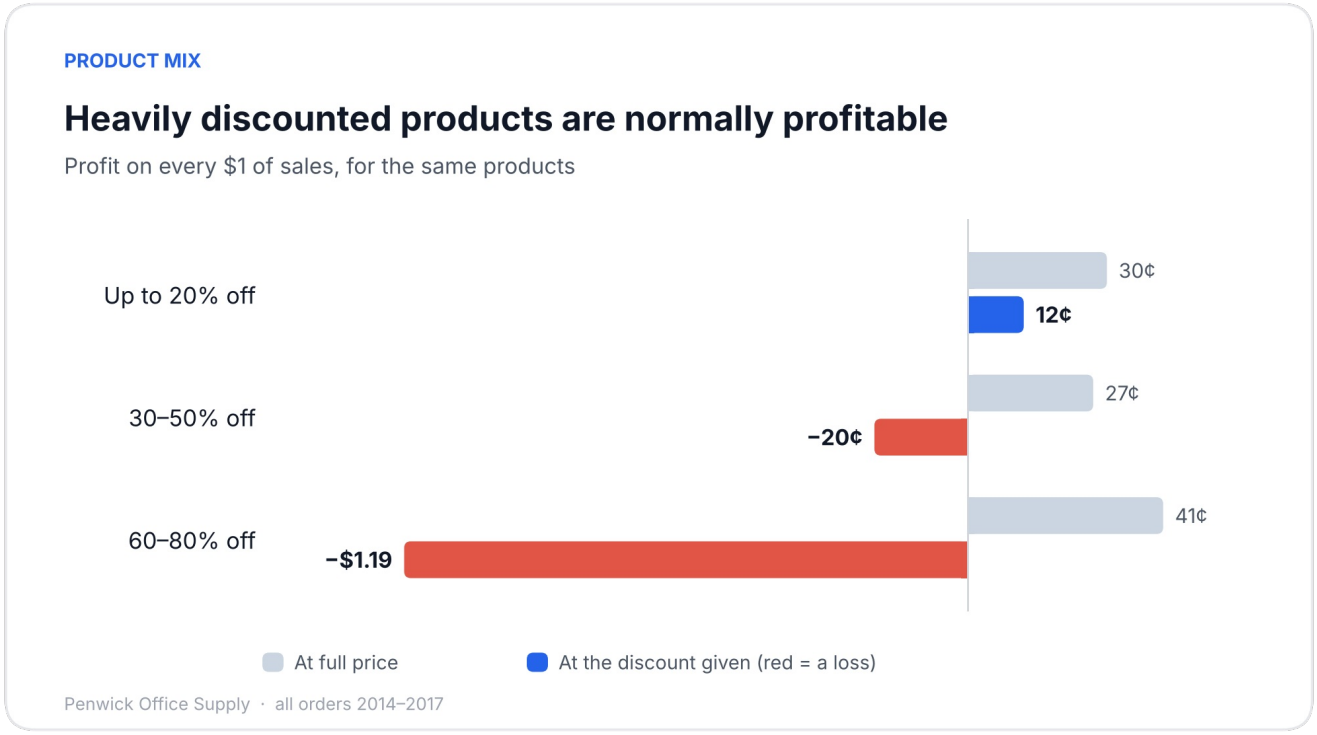


Band	Lines	Sales	Profit	Margin	Lines at a loss
No Discount	4,798	$1,087,908	$320,988	29.5%	0.0%
Standard (up to 20%)	3,803	$846,522	$100,785	11.9%	13.8%
Deep (over 20% to 50%)	537	$298,541	−$58,817	−19.7%	91.6%
Clearance (over 50%)	856	$64,229	−$76,559	−119.2%	100.0%

Source: `results/q1_band_summary.csv`

Furniture is hit hardest. A quarter of its sales (25.1%) were in the Deep band, at a −24.0% margin. Office Supplies never used the Deep band, but 5.5% of its sales went at 70–80% off, at −119.3%.

Q2. The losses come from the discount, not the product mix



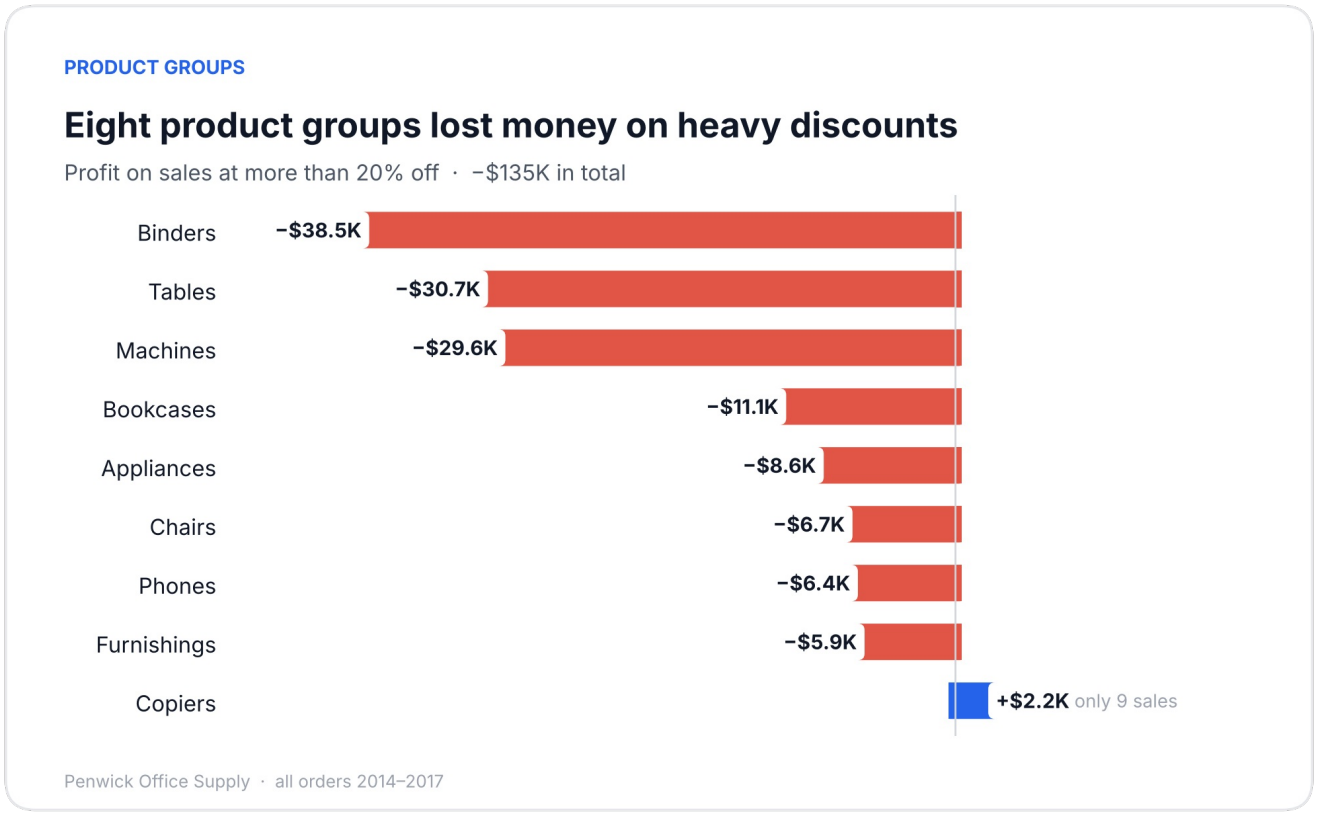
A band could look bad just because the products in it have low margins anyway. To rule that out, I compared each band with what its own products earn at no discount, sub-category by sub-category.

The Deep band's products would earn 26.6% at no discount, close to the company's 29.5%. They actually earned −19.7%. Product mix explains 2.9 points of that gap and the discount explains 46.3.

The Clearance band is mostly Binders, which make a 48% margin at full price. At clearance discounts the band earned −119.2%.

In the Standard band (up to 20%), margin drops by 17.6 points but stays positive at 11.9%.

Source: `results/q2_mix_vs_discount_effect.csv`



Looking inside the sub-categories:

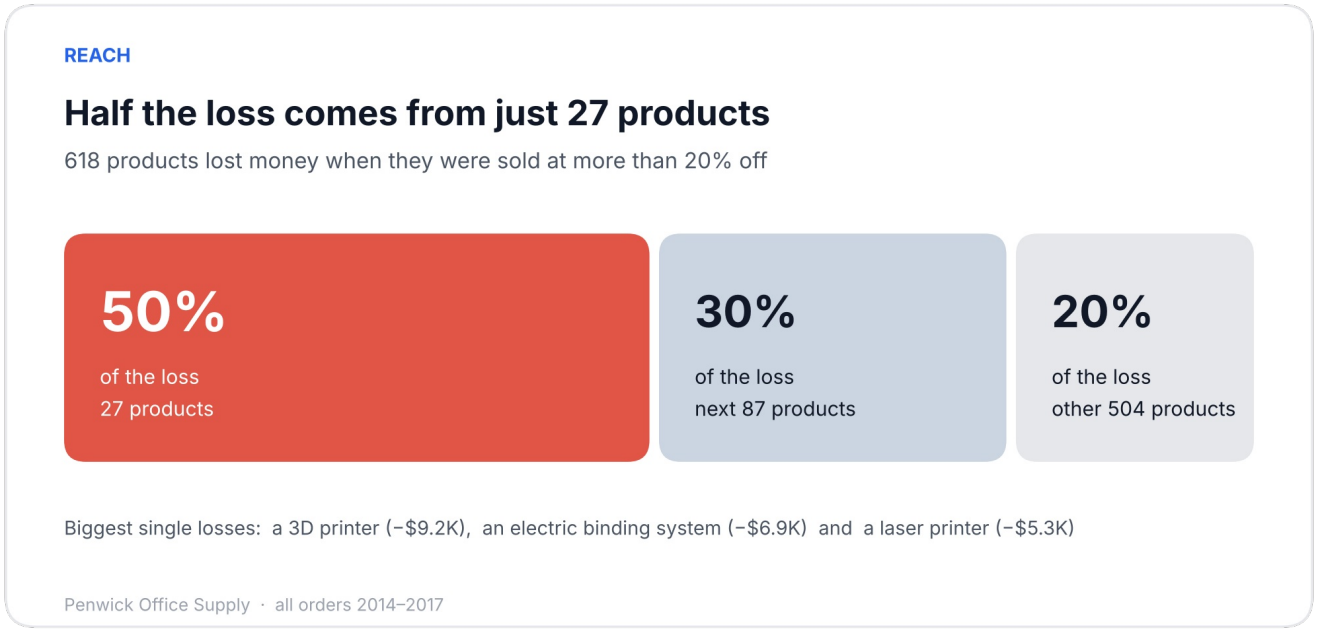
- Nine sub-categories sold lines above 20% off, and eight of them lost money. Copiers is the exception, with 9 lines at 40% off and a 12.9% margin, which is too few to rely on.
- Three sub-categories already lose money at up to 20%: Tables (−0.7%), Storage (−6.4%) and Supplies (−19.2%). A 20% cap won't fix them. That's a pricing problem, and Tables has no discount levels between 0% and 20% in the data, so I can't test a lower cap for it.
- Accessories, Art, Envelopes, Fasteners, Labels and Paper were never discounted above 20%, so the data says nothing about deeper discounts for them.

Source: `results/q2_subcategory_band_vs_baseline.csv`, `results/q2_subcategory_breakeven.csv`

Q3. Losses are spread across customers but concentrated in products

1,393 lines were discounted above 20%, and 1,348 of them (96.8%) lost money, a gross loss of $138,515. They touched 565 of 793 customers (71%), 1,020 of 5,009 orders (20%) and 649 products.

By sub-category, Binders (27.8%), Tables (22.2%) and Machines (21.7%) make up 71.7% of the loss. By product, 27 of the 618 loss-making products carry half of it and 114 carry 80%. The biggest are binding systems, 3D printers, laser printers and conference tables.



Entity	With a loss	Needed for 50% of loss	Needed for 80%
Products	618	27	114
Customers	558	41	146
Orders	998	55	213

Source: `results/q3_reach.csv`, `results/q3_loss_concentration_summary.csv`, `results/q3_product_loss_pareto.csv`

Deep discounting is routine, so a blanket rule makes more sense than reviewing deals one by one. The short product list is a good place to start the review.

Q4. What different caps would have meant (modelled)

These numbers come from a model. For each line above the cap, I undo its discount, apply the cap instead, and keep its implied cost. Nobody knows what share of that capped volume customers would still buy, so I ran it from 0% to 100%.



Cap	Lines capped	Profit change, none kept	Half kept	All kept
20%	1,393	+$135,376	+$177,284	+$219,191
30%	1,166	+$125,007	+$135,726	+$146,446
40%	933	+$99,559	+$94,856	+$90,152
50%	856	+$76,559	+$65,373	+$54,187

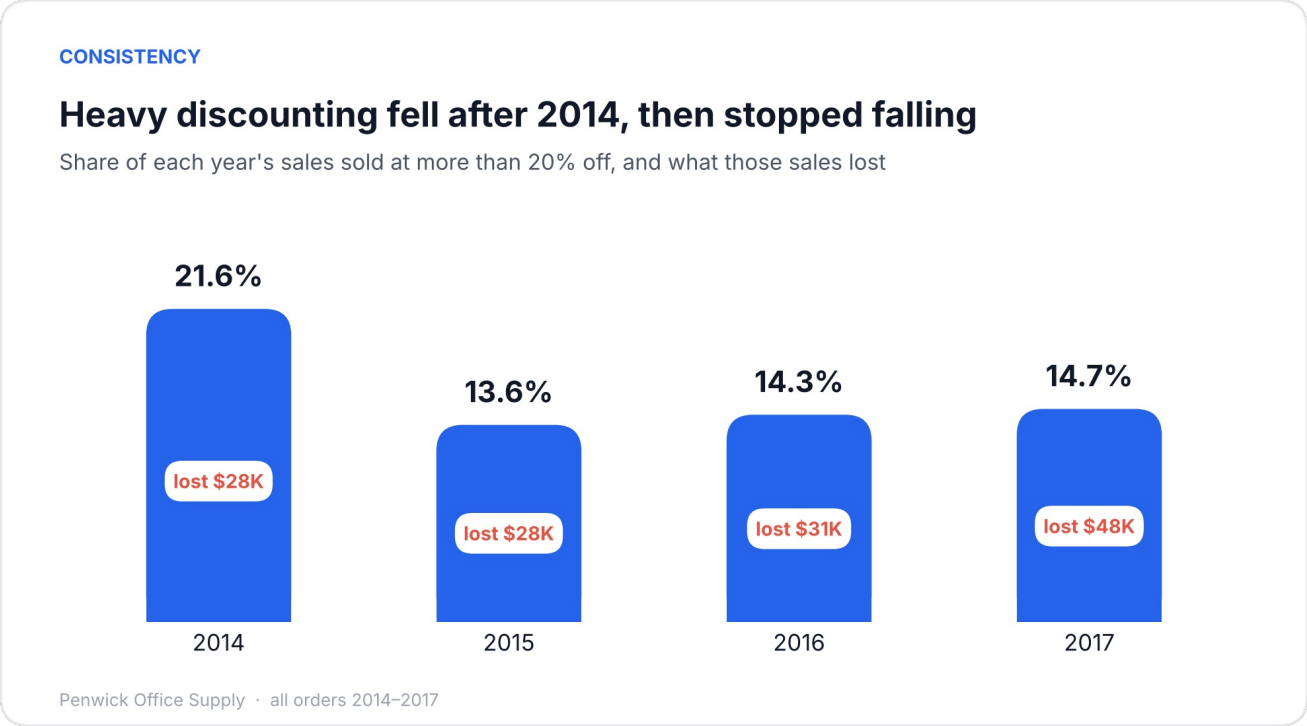
Source: `results/q4_scenarios_company.csv`

A 20% cap gives the biggest gain under every volume assumption. It is also the best cap for each of the eight loss-making sub-categories (`results/q4_best_ceiling_by_subcategory.csv`). Copiers only gains if at least half the volume is kept, and loses $2,183 if none is.

I also tried two harsher cases. In the first, customers drop the whole order whenever one line is capped. The other lines in those 1,020 orders take $24,099 of profit with them, and the 20% cap still comes out at **+$111,277** (`results/q4_whole_order_loss_case.csv`).

In the second, customers leave altogether. The 565 affected customers made $299,733 on their other lines, which is more than the company's total profit (`results/q4_customer_risk_case.csv`). With no capped volume kept, a 20% cap stops paying off if it costs more than about 45% of that ($135,376 ÷ $299,733). This is the risk the pilot has to measure.

Q5. The pattern held in every year, segment and region



The Deep band's margin was between −15.7% and −25.6% in every year, and between −15.4% and −24.9% in every region. The Clearance band sat between −98.6% and −136.9% everywhere.

The share of sales above 20% fell from 21.6% in 2014 to between 13.6% and 14.7% in 2015–2017, and then stopped falling.

Source: `results/q5_year_band.csv`, `results/q5_segment_band.csv`, `results/q5_region_band.csv`

The latest quarter shows where monitoring should start. In Q4-2017, 52.0% of Tables sales and 48.4% of Machines sales were discounted above 20%, and both lost money that quarter (`results/proc_last_6_quarters_subcategory.csv`).

5. Recommendations

Evidence	Action and owner	Expected benefit	Trade-off or dependency	Priority	How to evaluate
Q1, Q2, Q4	Cap discounts at 20% on Binders, Machines, Tables, Bookcases, Appliances, Chairs, Phones and Furnishings. Anything above needs Sales Manager approval. *Head of Sales*	Modelled +$111k to +$219k on 2014–17 orders. Not measured	Some customers may buy less or leave. Stock-clearance needs are unknown	1: largest, most consistent loss	Pilot (below)
Q2	Review list prices and costs for Tables, Storage and Supplies, which lose money even at 20% off. *Category / pricing manager*	Fixes losses a cap can't reach (Standard band: −$7,460 combined). Diagnostic only	Needs cost and price-list data this dataset doesn't have	2	Margin at 0–20% discount after the price change
Q3	Start with the 27 products behind half the deep-discount loss. Are they slow stock, mispriced, or discounted to win larger orders? *Category manager*	Explains why the discounts happen. Diagnostic	Needs reason codes or a stock-age report	2	Each product reviewed and classified
Q4	Run the cap as a pilot: two regions capped, two not, for two quarters. *Sales operations + analyst*	Turns the modelled range into a measured result	Regions differ slightly. Needs a baseline before the pilot	1: before full rollout	Compare margin, sales and repeat purchases by the 565 affected customers. Stop if their other-line profit falls by more than 45%
Q5	Track the share of sales above 20% and the margin per sub-category each quarter with `usp_LastQuartersSummary`. *BI analyst*	Shows whether the policy is followed	None	3	Report produced each quarter

6. Caveats

- The data is observational. Discounts go along with losses, but the data can't show what would have happened at a smaller discount. The pilot is how to find out.
- I don't know why each discount was given. If the deep-discounted items were old stock that would otherwise be written off, those discounts cost less than they look. The product review (recommendation 3) checks this.
- The model assumes each line's cost stays the same when its price changes. If list prices are inflated to leave room for discounts, the gains are overstated.
- Geography uses each customer's one listed location, not the delivery address, so the regional results are approximate.
- I excluded the sales-rep data. Reps are assigned per line rather than per order, and 2,288 orders have more than one rep. A rep's region doesn't match the customer's, and all nine reps have nearly identical product and discount mixes. Comparing them would mean nothing.
- One row looks like a duplicate (ROW_ID 3407 repeats 3406 apart from the rep). It changes total profit by less than 0.01%, so I kept it.
- This is a public teaching dataset. The patterns are real within it, but it isn't a live business.

7. Where to find things

Item	Location
Load script, fact view, analysis queries Q1–Q5	`sql/1_setup/00_load.sql`, `sql/1_setup/01_fact_view.sql`, `sql/2_analysis/10_q1…` to `sql/2_analysis/14_q5…`
What-if and monitoring stored procedures	`sql/3_procedures/20_scenario_procedure.sql`, `sql/3_procedures/30_monitoring_procedure.sql`
Validation (33 checks, all pass)	`sql/4_tests/90_tests.sql`, `scripts/cross_checks.py`, `results/validation_log.csv`, `docs/validation.md`
Data dictionary and metric definitions	`docs/data_dictionary.md`
Results behind every number	`results/*.csv`